The cost of computation and supercomputers

FLOPs, GPU hours, dollars and energy

Carlos Cotrini

The same three questions, asked for the AI

Was kann *es* wissen? → Weekend 5: RAG

Was soll *es* tun? → Weekend 6: Agentic AI

→

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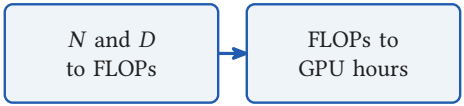
Weekend 5 was what it can know, Weekend 6 what it should do. And this weekend is what it may become.

What the next hour computes

N and D
to FLOPs

$6ND$

What the next hour computes



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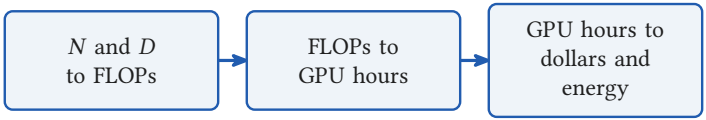

Section 1

Dividing a computation, 1793

1

Adam Smith, 1776

The pin maker's workshop. *Epinglier*, Diderot and d'Alembert's *Encyclopedie*, 1762

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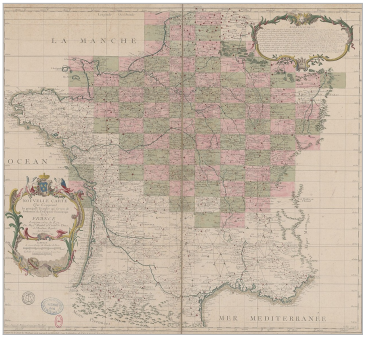
- *The Wealth of Nations*, Book I, chapter 1: **Of the Division of Labour**
- One workman making a whole pin: perhaps twenty in a day, perhaps not one
- Ten men, split across eighteen operations: forty eight thousand
- Same people, same tools. Only the organisation changed

Dividing a computation, 1793

The task: the area of every parcel in France



The principal triangles of France. Maraldi and Cassini de Thury, 1744

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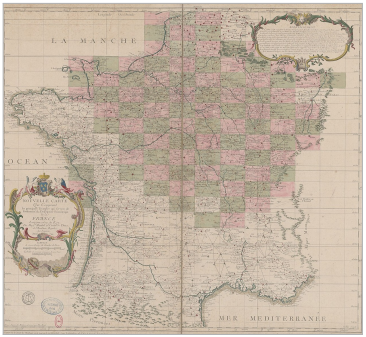

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Dividing a computation, 1793

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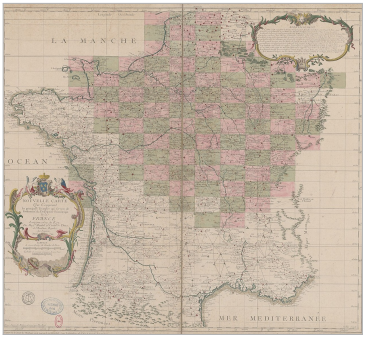

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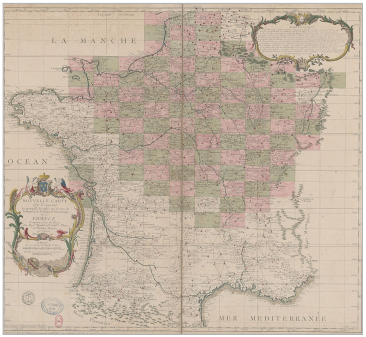

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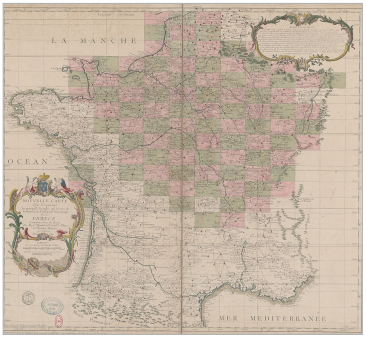

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- The Revolution replaces tax privilege with one tax on land
- So every parcel has to be measured, and its area computed
- Gaspard de Prony directs the Bureau du Cadastre, which does the computing
- Land is measured in triangles, because angles are quick and chains are slow

Dividing a computation, 1793

Three tiers, and what each hands down

Five or six mathematicians
Choose the formulas and the blocks

A computing room, Harvard College Observatory, about 1890

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Seven or eight analysts
Per block, one exact value
and its differences

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About twenty computers
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Je conçus tout à coup l'idée d'appliquer la même méthode à l'immense travail dont je m'étais laissé imposer le fardeau, et de fabriquer mes logarithmes comme on fabrique des épingles.

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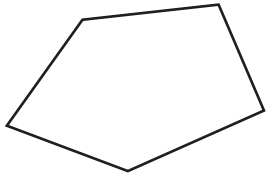
Manufacturing my logarithms as one manufactures pins.
Gaspard de Prony, Notices, 1824, p. 5.

Dividing a computation, 1793

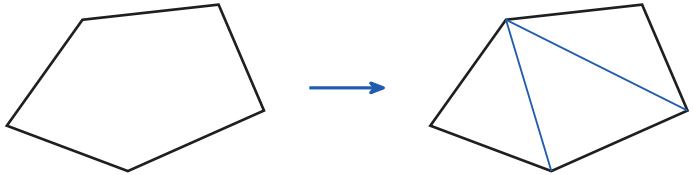
One parcel, many triangles, one each



One parcel

Dividing a computation, 1793

One parcel, many triangles, one each

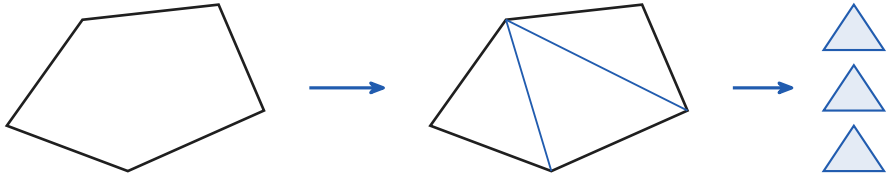


One parcel

Cut into triangles

Dividing a computation, 1793

One parcel, many triangles, one each



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Dividing a computation, 1793

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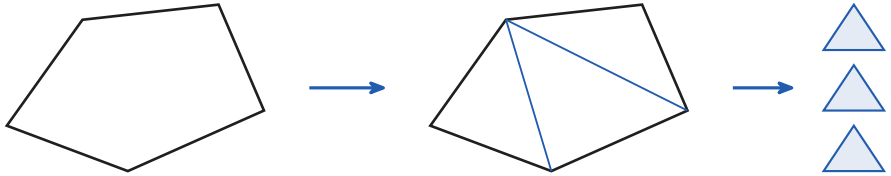
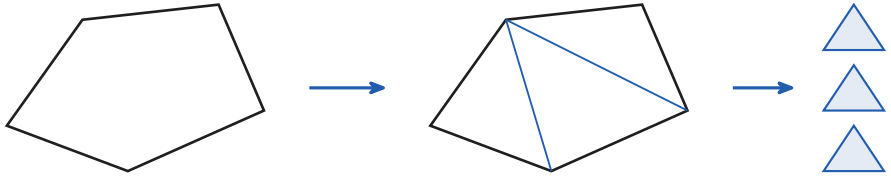


- A clerk never sees a parcel. He is handed one triangle and returns one area

Dividing a computation, 1793

One parcel, many triangles, one each



- A clerk never sees a parcel. He is handed one triangle and returns one area
- Which is why twenty of them can work at once without talking to each other

One side, two angles, and a table of logarithms

$$\text{Area} = \frac{c^2 \, \sin A \, \sin B}{2 \sin C}$$

400 m

Measured with the chain

One side, two angles, and a table of logarithms

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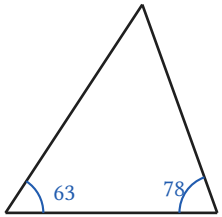

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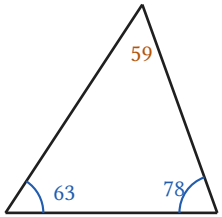

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The third angle is free:
200 − 63 − 78

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Dividing a computation, 1793

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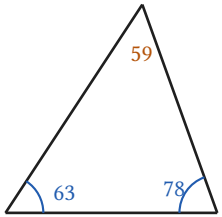

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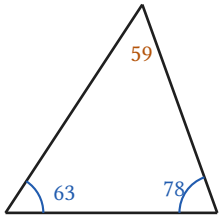
By hand that is a square, two products and a division

One side, two angles, and a table of logarithms



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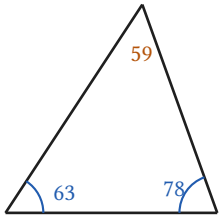
Dividing a computation, 1793

One side, two angles, and a table of logarithms



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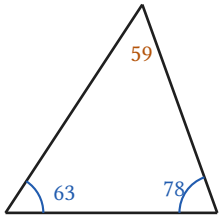
	$2\log 400$	$+5.2041200$
$+$	$\log\sin 63$	-0.0778938
$+$	$\log\sin 78$	-0.0264654
$-$	$\log\sin 59$	-0.0970812
$-$	$\log 2$	$+0.3010300$

Five lookups in a seven figure table, then four additions. The parcel is ours; the units and the method are the period's.

Dividing a computation, 1793

One side, two angles, and a table of logarithms



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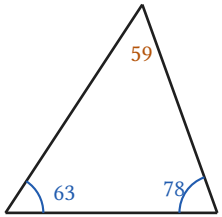

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	Area	$78\,670\ \text{m}^2$

about 7.9 ha

Five lookups in a seven figure table, then four additions. The parcel is ours; the units and the method are the period's.

Section 2

One operation, and who does it

2

A FLOP is one operation on two numbers

One multiply accumulate

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One multiply accumulate

$a \times b$
one multiply

One operation, and who does it

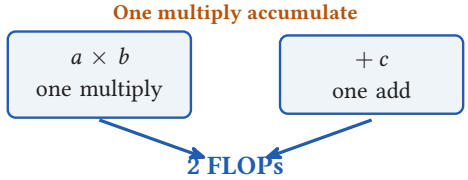
A FLOP is one operation on two numbers

One multiply accumulate

$a \times b$
one multiply

$+ c$
one add

A FLOP is one operation on two numbers



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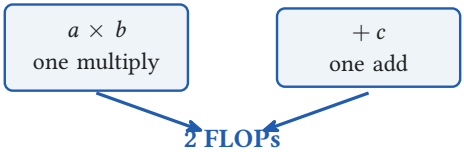

Count, or rate

FLOP: a count of operations
how much work

A FLOP is one operation on two numbers

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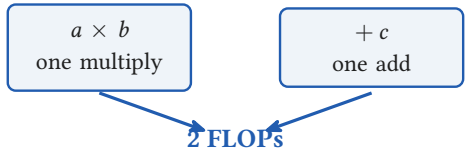


Count, or rate

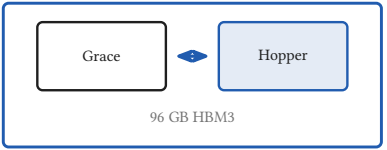
FLOP: a count of operations
how much work

FLOP/s: operations per second
how fast

One operation, and who does it

One computer, then and now

A computer room, NACA, 1949

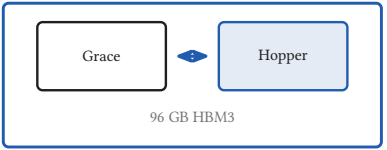


One GH200. Alps holds 10,752

Human computer, 1949	One GH200, 2024

One computer, then and now

A computer room, NACA, 1949



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	Human computer, 1949	One GH200, 2024
Rate	1,000 additions a day	990 TFLOP/s dense BF16

One operation, and who does it

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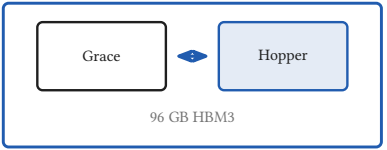

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Rate	1,000 additions a day	990 TFLOP/s dense BF16
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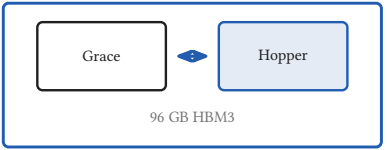

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	Human computer, 1949	One GH200, 2024
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One operation, and who does it

One computer, then and now

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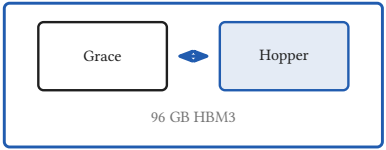

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	what one person eats	what six or seven people eat

One operation, and who does it

One computer, then and now

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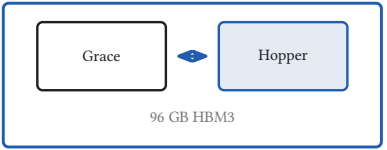

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- The clerk's rate is what the shop *delivered*. The chip's is what it *could* do

One operation, and who does it

A GH200 is a CPU and a GPU on one module

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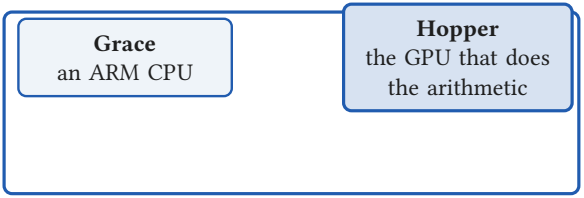

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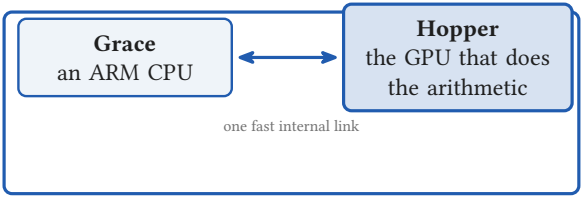

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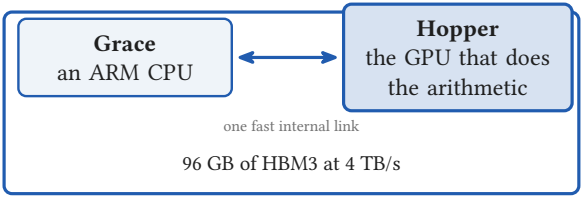

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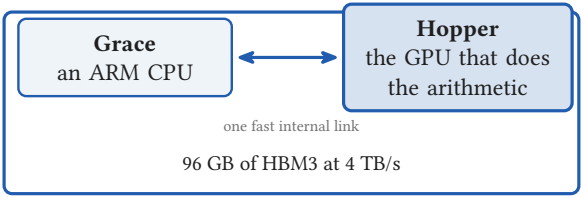

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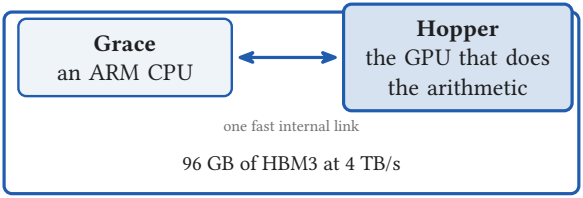

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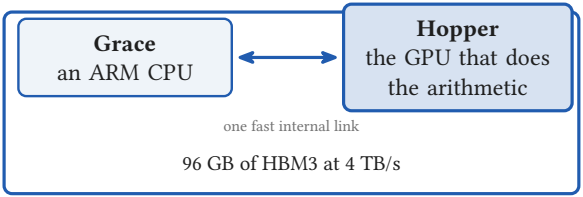

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- Between them 900 GB/s, about seven times what a plug-in card gets

Alps is an HPE Cray EX254n

The TOP500 entry reads: *Alps, HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS*

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HPE Cray EX
the supercomputer platform:
cabinets, liquid cooling,
power, the fabric

One operation, and who does it

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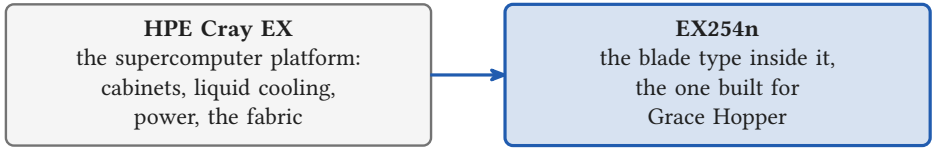

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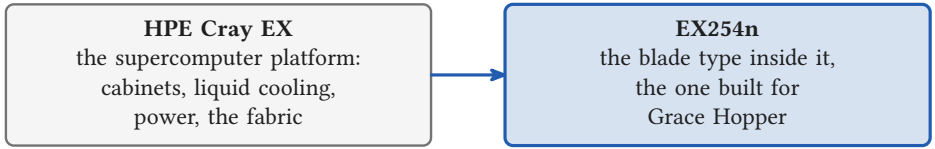

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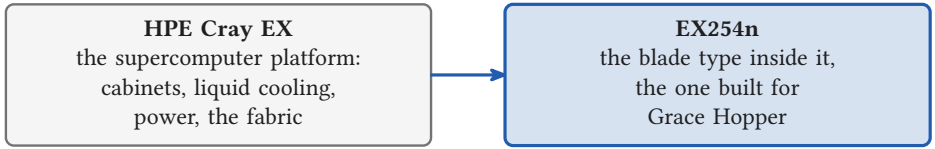

The trailing n marks the NVIDIA accelerated blades

- The blade choice is what makes Alps a machine for AI rather than a conventional CPU cluster

One operation, and who does it

The atelier against Alps

A collection of computers	Rate	Power
Prony’s atelier, 1795 twenty people	0.694 additions/s	about 2 kW

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Fifteen billion billion times the rate, on three and a half thousand times the power

One operation, and who does it

What the TOP500 actually measures

A list of the 500 most powerful known supercomputers, published every June and November since 1993. Alps is rank 10 in June 2026.

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So even on its own benchmark it reaches 75.7 percent, and drew 7.1 MW doing it

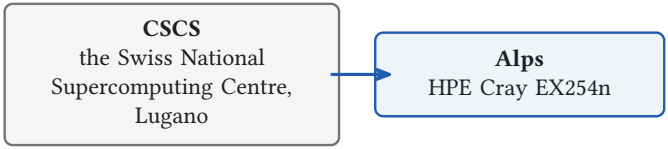
One operation, and who does it

CSCS and Alps, the machine behind the anchor

CSCS
the Swiss National Supercomputing Centre, Lugano

One operation, and who does it

CSCS and Alps, the machine behind the anchor



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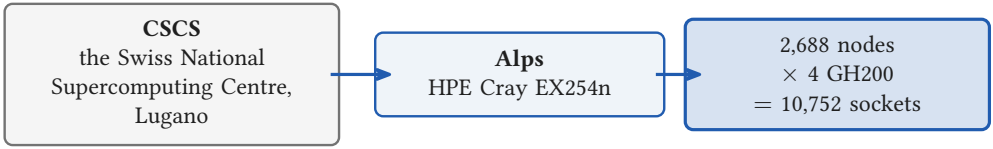

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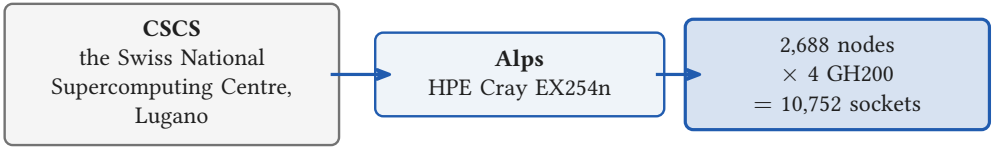

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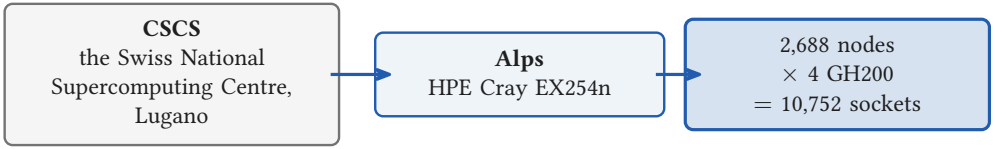

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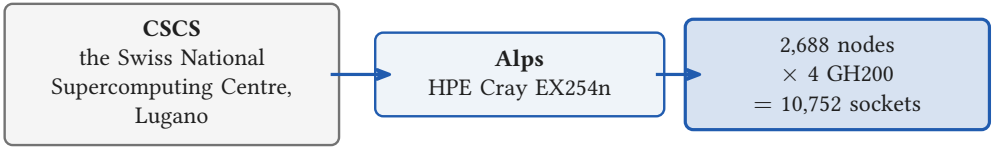

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Machine: top500.org/system/180259 and cscs.ch/computers/alps.

One operation, and who does it

One second of a GH200, in three registers

The same one second of arithmetic

The mean age of all 8.2 billion people alive, one addition each

8.3 microseconds

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ResNet-50: He et al., arXiv:1512.03385, 3.8 billion multiply-adds, so 7.6 GFLOP per image. A peak count.

One operation, and who does it

One training step of linear regression

1) Data: price and d features

	Price	Size	Rooms		Age
House 1	y_1	x_{11}	x_{12}	$\cdots$	x_{1d}
House 2	y_2	x_{21}	x_{22}	$\cdots$	x_{2d}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
House n	y_n	x_{n1}	x_{n2}	$\cdots$	x_{nd}

One operation, and who does it

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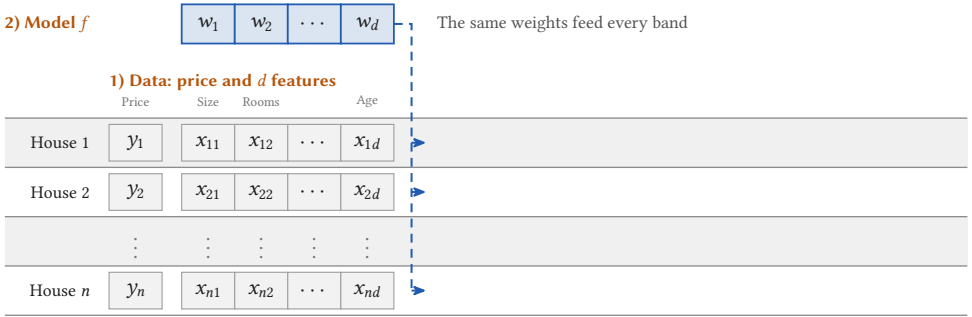

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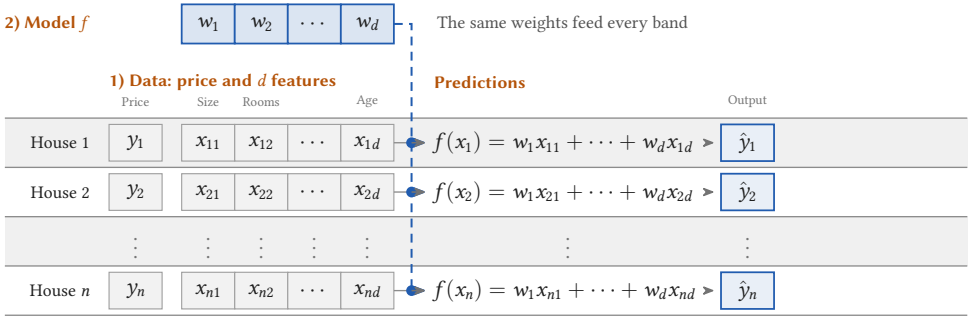

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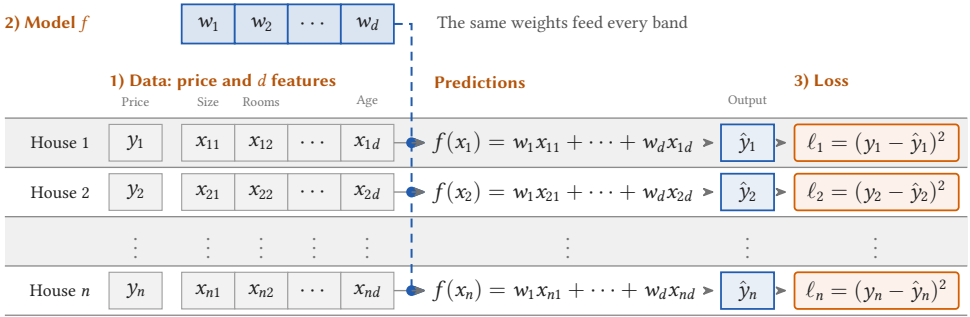

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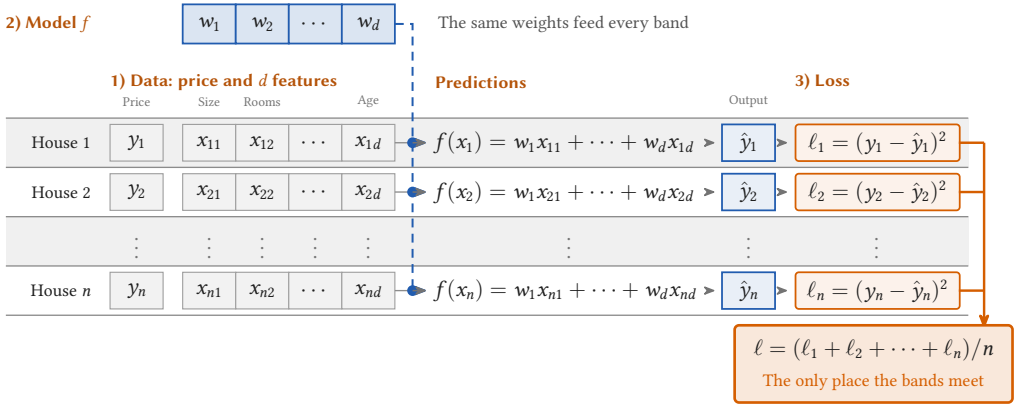

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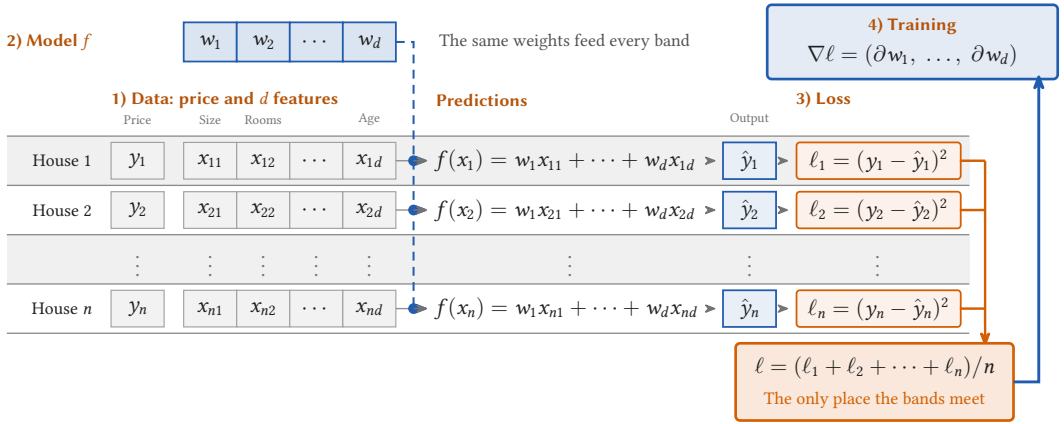

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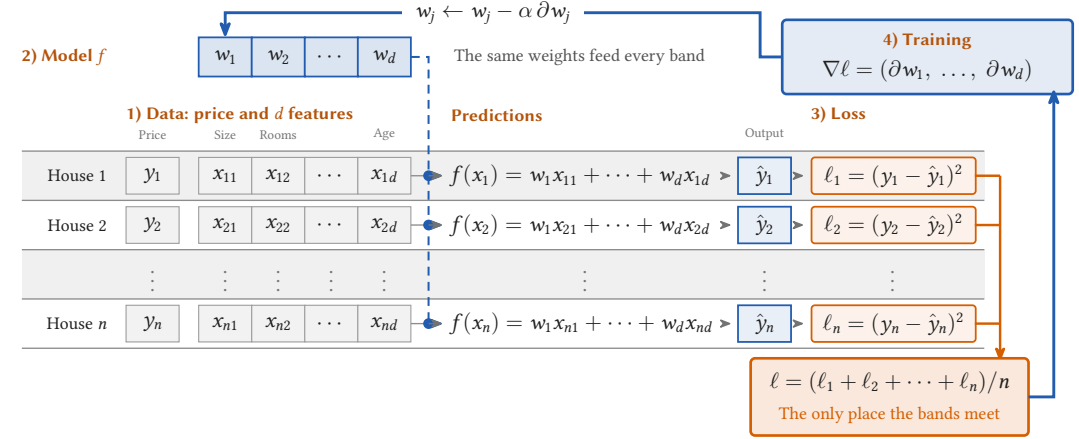

Section 3

What the headline rate hides

3

One floating point number, taken apart

Start with

$$\pi = 3.14159265\ldots$$

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A mantissa times a power of two

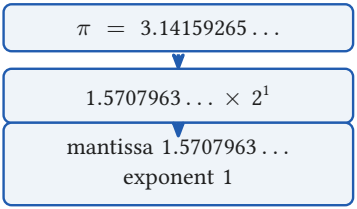
$$1.5707963\ldots \times 2^{1}$$

One floating point number, taken apart

Start with

A mantissa times a power of two

So really two numbers



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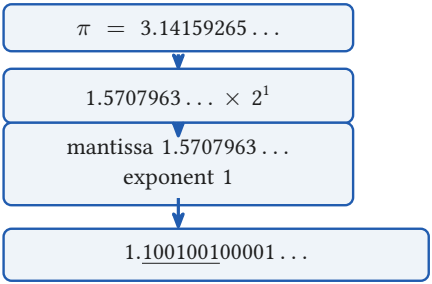
The mantissa in binary



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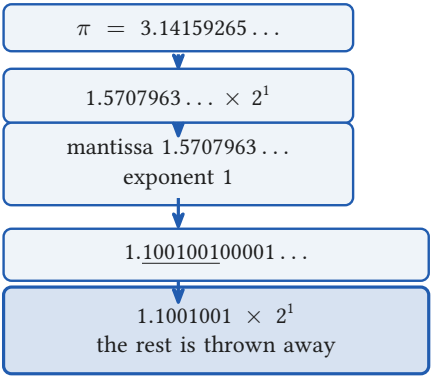
The mantissa in binary

bf16 keeps seven of those bits



What the seven bits cost

The sixteen bits

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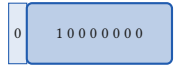

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Back to decimal
3.140625

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What the seven bits cost

The sixteen bits

Back to decimal	We started from
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Lost 0.00097, or 0.03 percent

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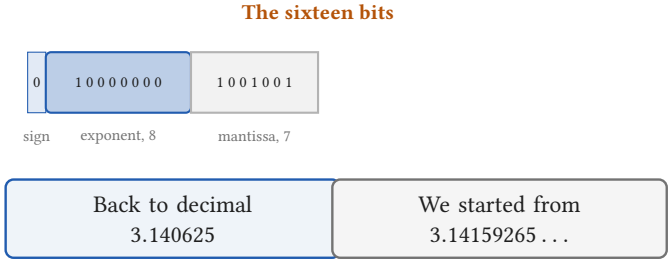

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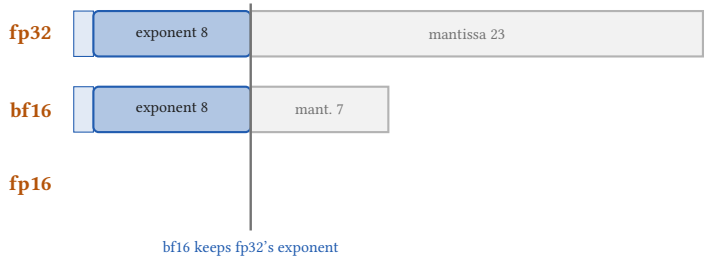
- Seven bits cannot hold π, so the number stored is not the number you asked for

Not every FLOP costs the same

What the headline rate hides

Not every FLOP costs the same



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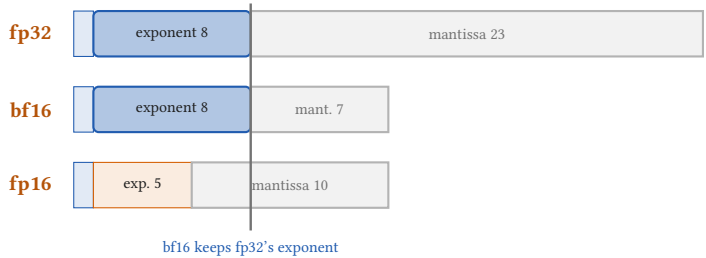

Why we say dense BF16, and not just BF16

The datasheet's fast number assumes a pattern

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1,979 TFLOP/s

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- Sparsity is just something you impose on a finished model

Capacity and bandwidth are different numbers

96 GB
how much fits

What the headline rate hides

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4 TB/s
how fast you can move it

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96 GB
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4 TB/s
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Decides how quickly the arithmetic units can be fed once the data is there

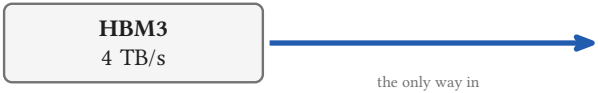
- 96 GB is less than an ordinary server holds. That is the trade: HBM is stacked on the package and wired short and wide, which is what buys the speed

Two hundred and forty six operations per byte

HBM3
4 TB/s

What the headline rate hides

Two hundred and forty six operations per byte



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A rate over a rate: the seconds cancel, and what is left is operations per byte

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So unless the computation reuses each byte about that many times, the units wait. A large matrix multiply does reuse. Many other operations do not.

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- This is one of the reasons a real run reaches 21 to 46 percent of peak

Section 4

The machine, and what it draws

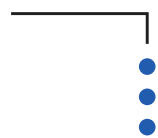
4

A watt is a rate, a kilowatt hour is an amount

660 W

How fast energy arrives

A watt is a rate, a kilowatt hour is an amount



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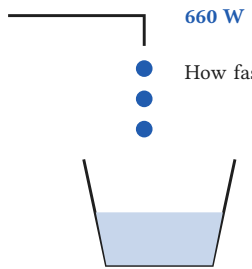

660 W

How fast energy arrives

0.66 kWh after one hour

How much arrived

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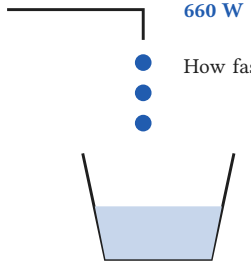

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How fast energy arrives

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How much arrived

Watch the slash

km/h the slash divides, so it is a speed

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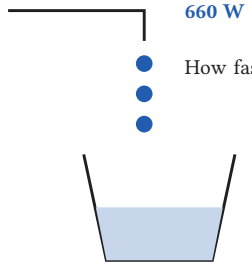

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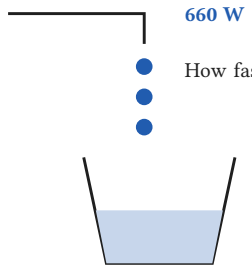

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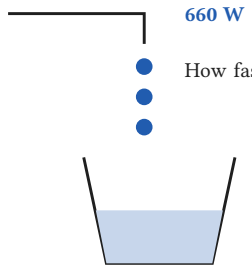

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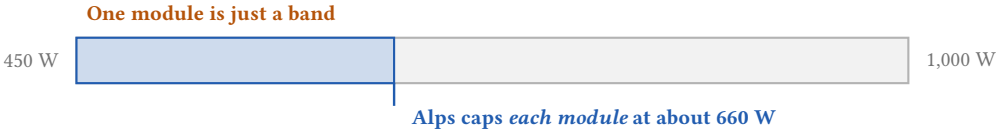
- Rate × time = amount, twice over: FLOP/s × seconds = FLOPs, and watts × hours = kWh

A wattage becomes a bill in two multiplications

One module is just a band

A wattage becomes a bill in two multiplications



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Alps caps *each module* at about 660 W

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for one hour

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× time

= 0.66 kWh
of energy

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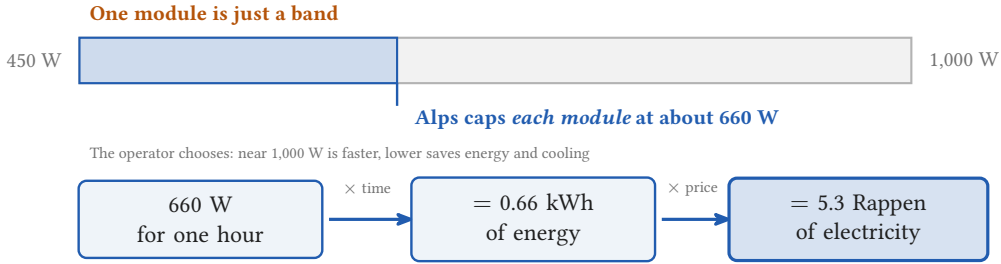

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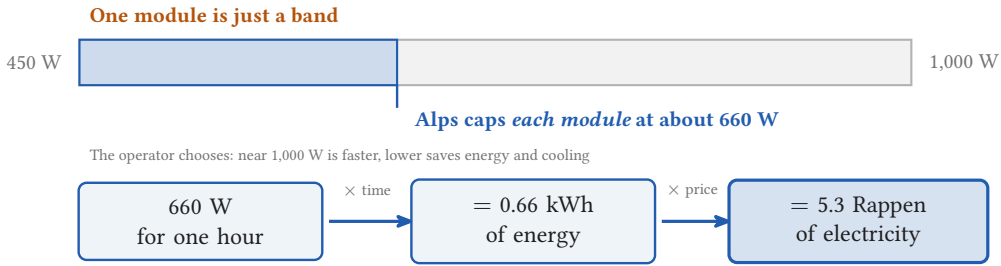

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- That is one module. All 10,752 is 7.1 megawatts, about 570 francs an hour

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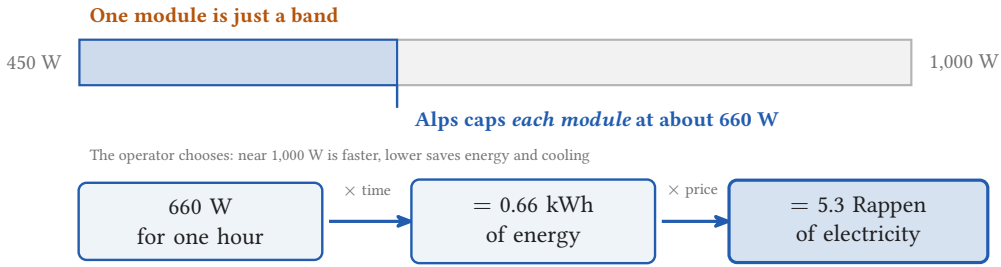

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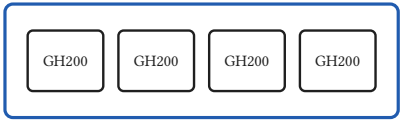
- That is one module. All 10,752 is 7.1 megawatts, about 570 francs an hour
- The whole Apertus run is 3.96 GWh, about 0.3 million francs

Nodes and sockets

One node

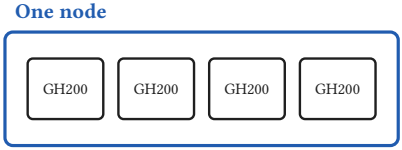
Nodes and sockets

One node



One operating system, one address space

Nodes and sockets



Socket one processor module
Node one independent computer

Nodes and sockets

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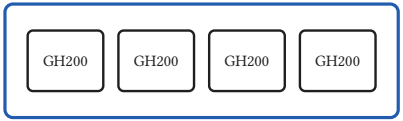

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2,688 nodes × 4 modules = 10,752 sockets

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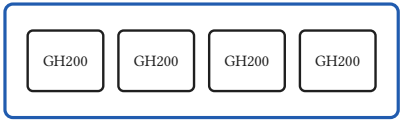

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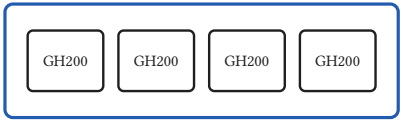

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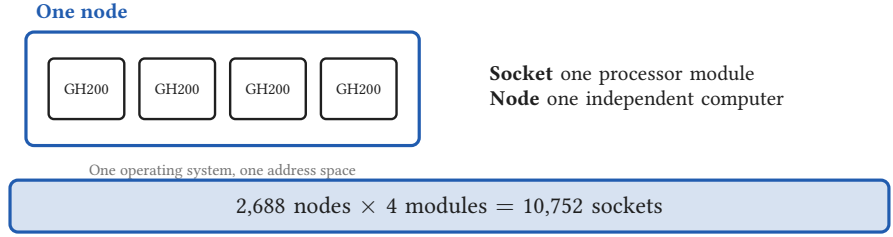
2,688 nodes × 4 modules = 10,752 sockets

Socket to socket, inside the node: 150 GB/s

Between nodes: 25 GB/s per GPU, six times less

Nodes and sockets



- The node boundary is where the bandwidth falls off

Four links, and each step down costs you

- **4,023 GB/s** the GPU to its own HBM3

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Each step down costs about a factor of five. Top to bottom, 161 times.

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Each step down costs about a factor of five. Top to bottom, 161 times.

- So you keep the data at the fast end and cross the slow links as rarely as you can

Section 5

From a model to GPU hours

5

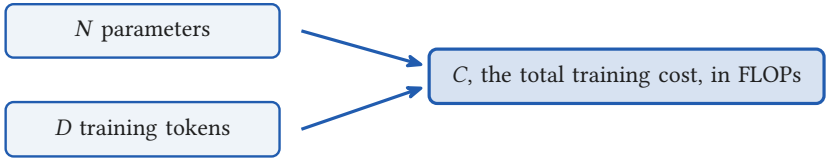
From a model to GPU hours

Two numbers price a training run

N parameters

D training tokens

Two numbers price a training run



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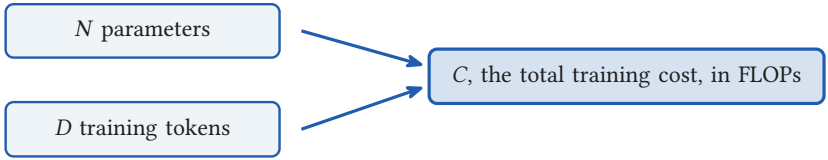

$$C = \underbrace{N}_{\text{parameters}}$$

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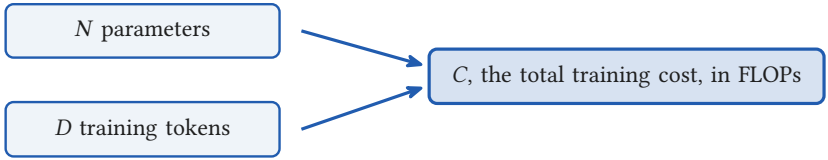

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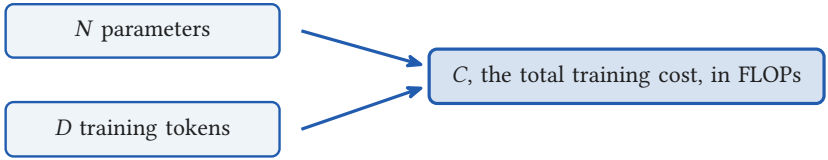

$$C = \underbrace{6}_{\text{per parameter, per token}} \times \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{tokens}}$$

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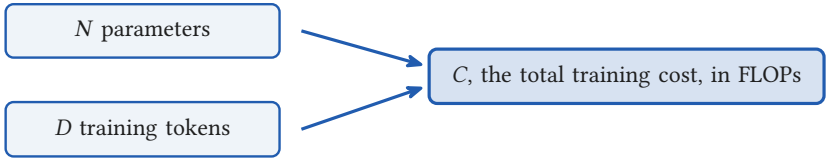

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Kaplan et al. 2020, section 2.1. Where the 6 comes from is the 12:00 lecture; for the next fifty minutes it is a rule of thumb.

A million by a million is two minutes

$N = 10^6$ parameters

$D = 10^6$ tokens

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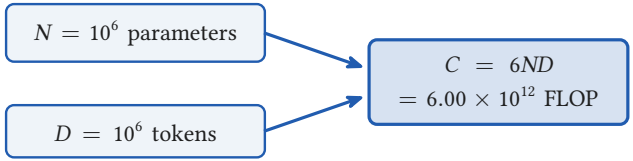

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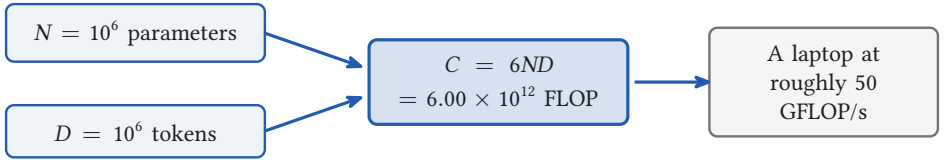

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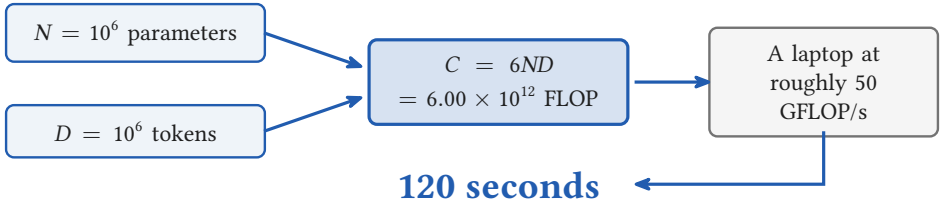

The anchor: Apertus, trained on Alps

Apertus-70B
Released 2 September 2025, Apache 2.0
70.60×10^9 parameters, 15 trillion tokens
1,811 languages, 40 percent of it not English

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Model: arXiv:2509.14233. Run: arXiv:2604.12973. Machine: top500.org/system/180259.

From a model to GPU hours

The same formula on Apertus-70B

$$N = 70 \times 10^{9}$$

$$D = 15 \times 10^{12}$$

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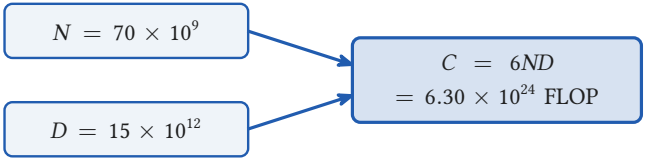

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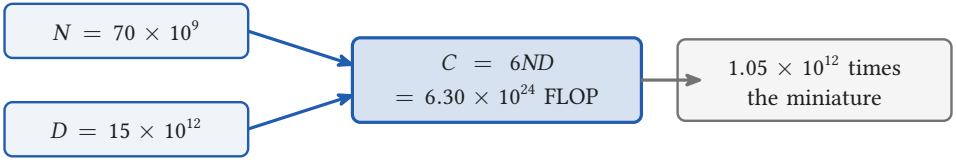

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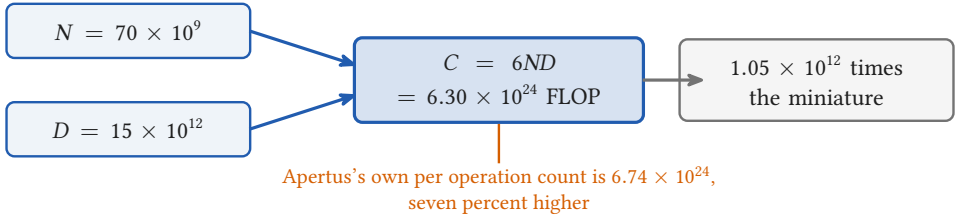

From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion tokens, on up to 4,096 GPUs.

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6 million GPU hours × 660 W = **3.96 GWh** × 8 Rappen = **0.32 million francs**

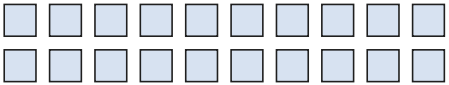
Section 6

Peak, and what a real run gets

6

Peak, and what a real run gets

Counting the workshop is not measuring it



Twenty computers, one desk each

Peak, and what a real run gets

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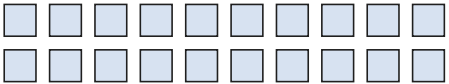

Twenty computers, one desk each

Count the desks, multiply by how fast one clerk works

Peak, and what a real run gets

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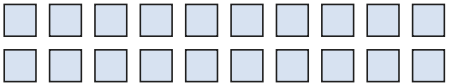

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That is the workshop's **peak**

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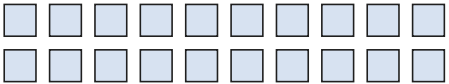

Twenty computers, one desk each

Counted from the doorstep. Nothing was run.

Count the desks, multiply by how fast one clerk works

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Peak, and what a real run gets

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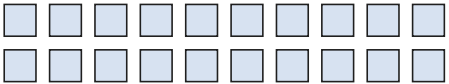


Twenty computers, one desk each

Counted from the doorstep. Nothing was run.

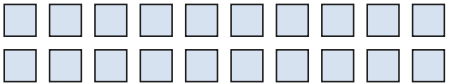
Now watch for an hour

Count the desks, multiply by how fast one clerk works

That is the workshop's **peak**

Peak, and what a real run gets

Counting the workshop is not measuring it



Twenty computers, one desk each

Counted from the doorstep. Nothing was run.

Now watch for an hour

A clerk waits for the next sheet.
Two clerks stop to compare a figure.
The batch is small, so some desks are empty.

Count the desks, multiply by how fast one clerk works

↓

That is the workshop's **peak**

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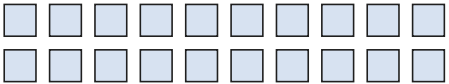

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That is the workshop's **peak**

- The clerks were at full speed the whole time. They were not always given something to do

Peak is just a ceiling

One GH200, one second

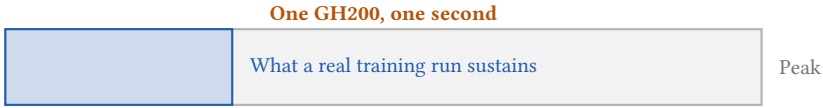
Peak

Peak is just a ceiling

every arithmetic unit busy, every cycle, data always ready

Peak, and what a real run gets

Peak is just a ceiling

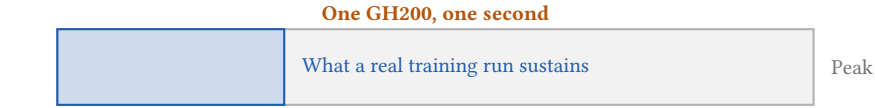


every arithmetic unit busy, every cycle, data always ready

A whole training run, averaged, drawn on one second so you can see it

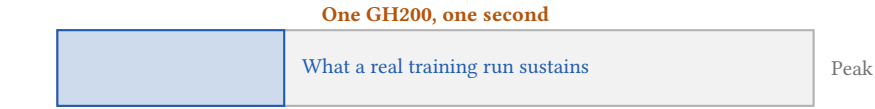
Peak, and what a real run gets

Peak is just a ceiling



The gap is waiting on memory, talking to other chips, and kernels that do not fill the hardware.

Peak is just a ceiling



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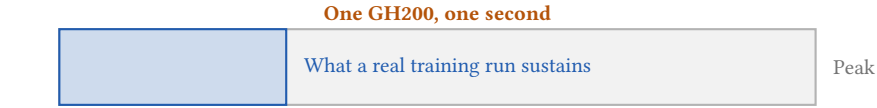

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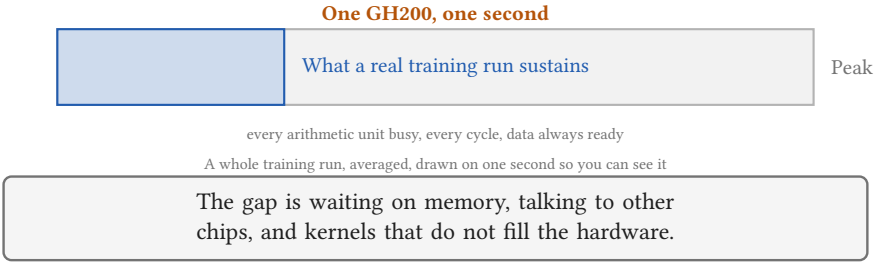

- Nobody reaches peak. The datasheet number is a ceiling
- The share you do reach is model FLOPs utilisation, written MFU
- It is the blue bar over the whole bar, so MFU 0.30 means thirty percent of peak

Peak, and what a real run gets

Published runs reach 21 to 46 percent of peak

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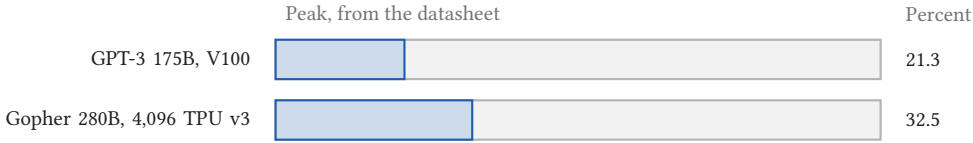

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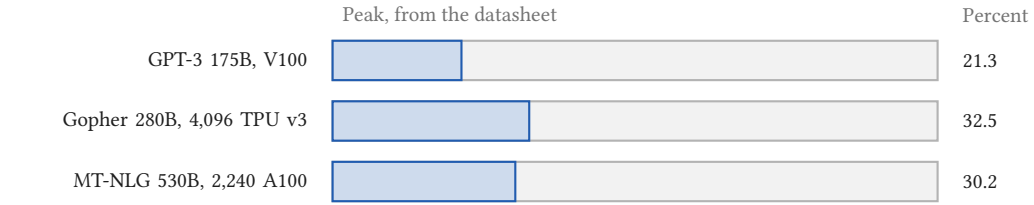

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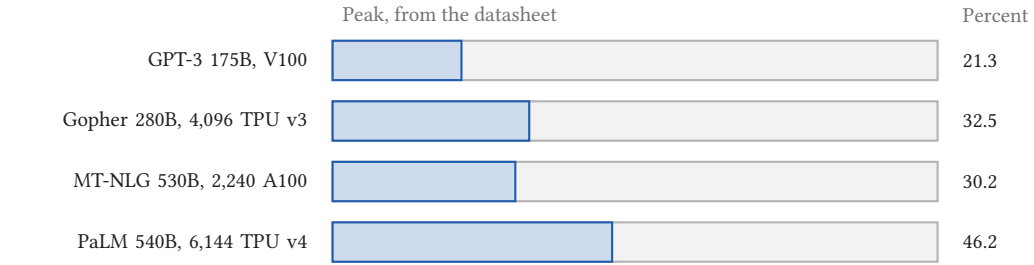

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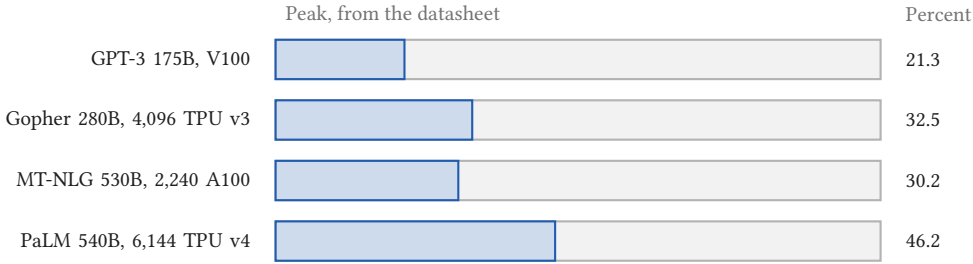

- Two ran on TPUs, so the umbrella word is accelerator: GPU, TPU, or GH200

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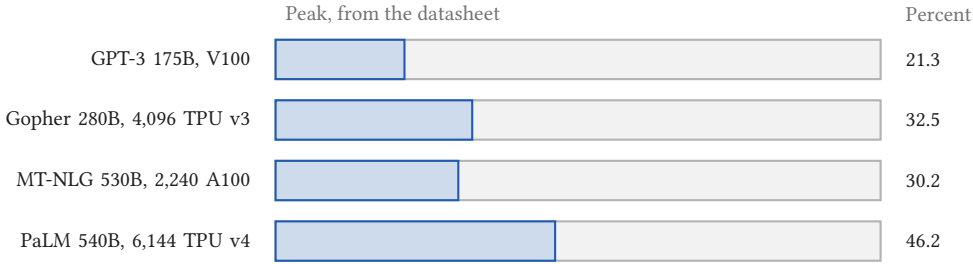

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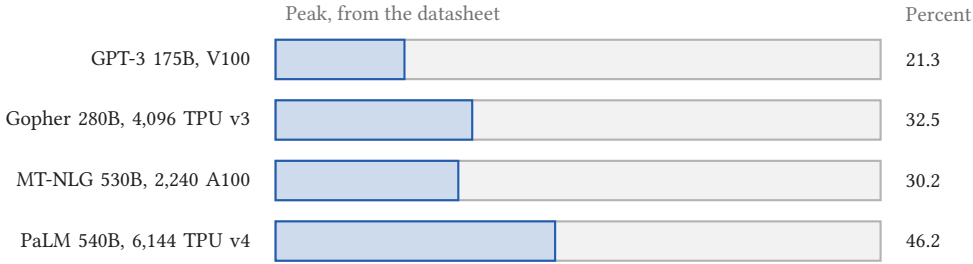

- Two ran on TPUs, so the umbrella word is accelerator: GPU, TPU, or GH200
- Model FLOPs utilisation: the share of peak a training run reaches
- Assuming 100 percent is wrong by a factor of two to five

Where the utilisation formula comes from

$$\text{MFU} = \frac{\text{observed throughput}}{\text{theoretical maximum}}$$

Where the utilisation formula comes from

$$\text{MFU} = \frac{\text{tokens/s} \times \text{FLOP per token}}{\text{peak FLOP/s} \times \text{chips}}$$

- Tokens per second is what you observe. FLOP per token is what you count

Where the utilisation formula comes from

$$\text{MFU} = \frac{\frac{D}{T} \times 6N}{\text{peak FLOP/s} \times \text{chips}}$$

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- So the measurement enters through the hours, and $6ND$ supplies the work

Peak, and what a real run gets

Six million hours, measured against peak

The consumption
Over 6 million GPU hours, the 70B production run

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analytic

$6ND$ for the 70B

$$\frac{6.30 \times 10^{24}}{2.14 \times 10^{25}} = 29.5 \textbf{ percent}$$

Section 7

Dollars and energy

7

Dollars and energy

Six million hours, priced at cloud rates

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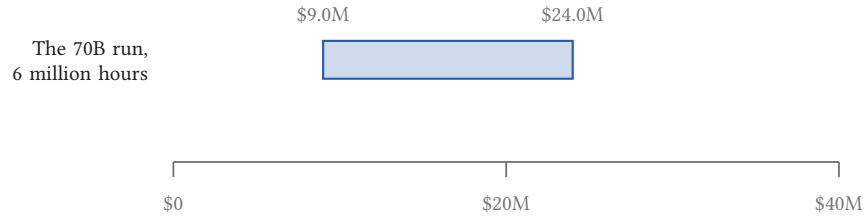

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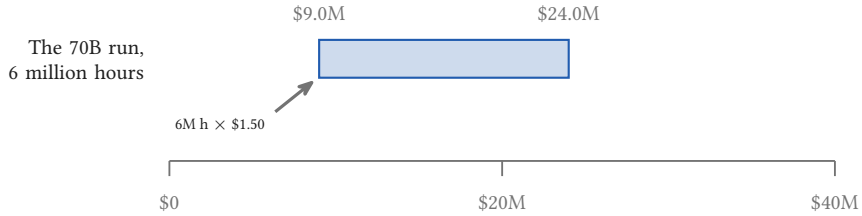

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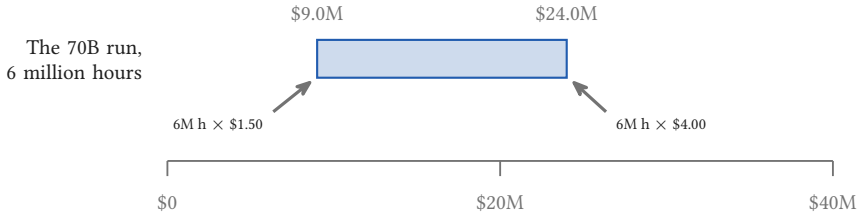
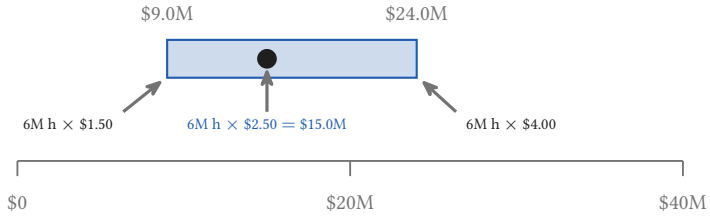


Dollars and energy

Six million hours, priced at cloud rates

The 70B run,
6 million hours



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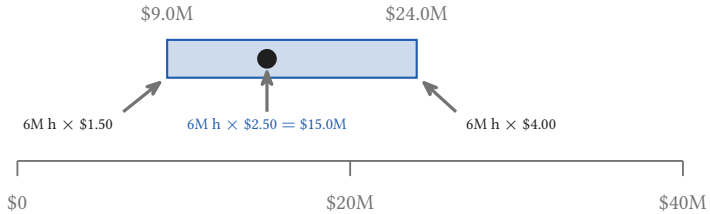

The $1.50 to $4.00 band is an H100 rental survey, November 2025.

Dollars and energy

The same hours, in gigawatt hours

8.85 million GPU hours
(90 days × 4,096 GPUs)

Dollars and energy

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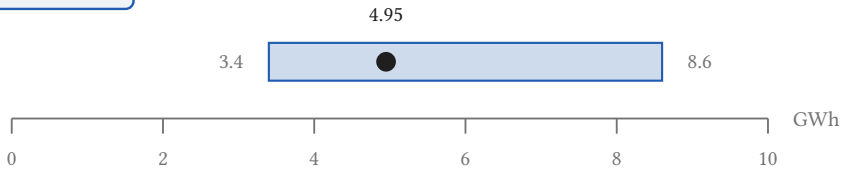

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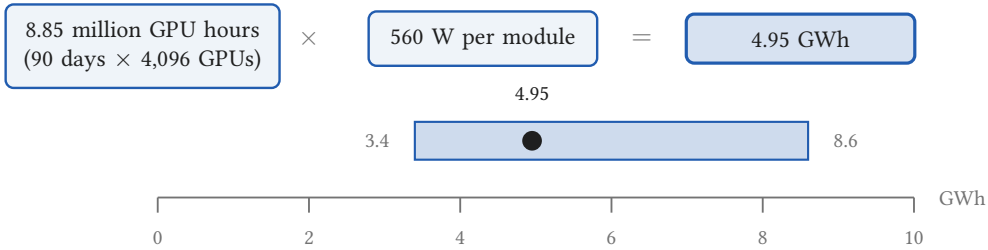

- About 1,100 Swiss households for a year, at 4,500 kWh each

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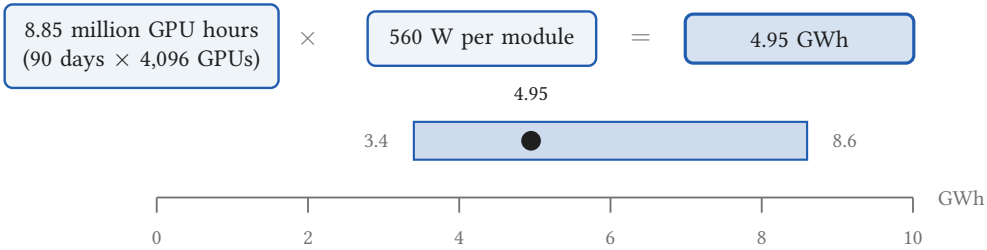

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Range 750 to 1,900 households, over four sourced module figures: **560 W** tech report, arXiv:2509.14233; **660 W** CSCS cap, docs.cscs.ch/running/slurm; **855 W** from the TOP500 run, top500.org/system/180259; datasheet band 450 to 1,000 W.

Dollars and energy

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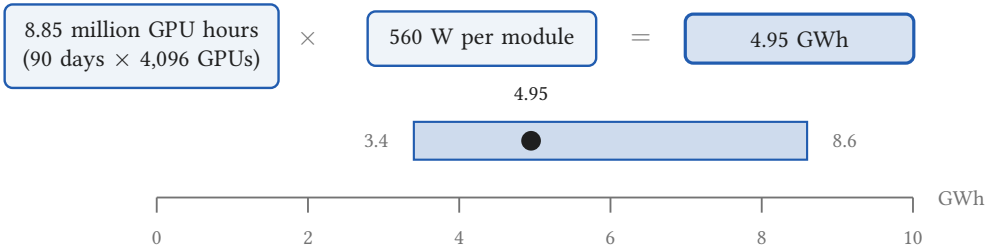

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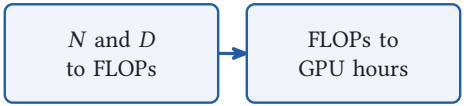
Household 4,500 kWh a year is ElCom's H4 profile. CSCS publish a PUE below 1.2, cooling from Lake Lugano (cscs.ch).

What you can price

N and D to FLOPs

$$6ND$$

What you can price



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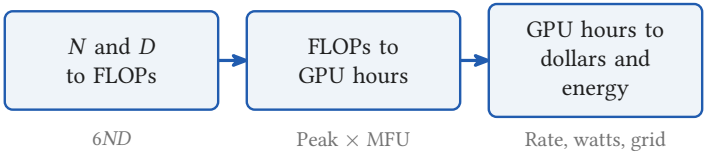

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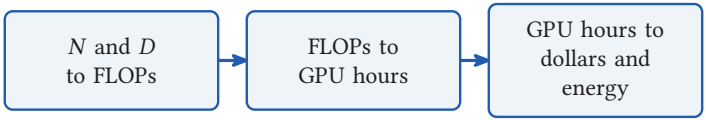

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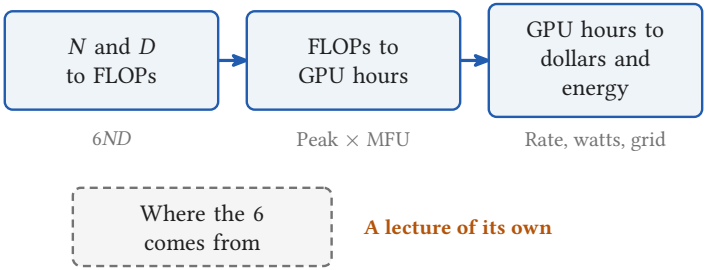
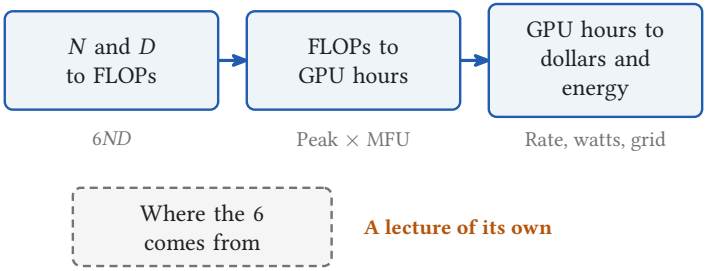


- You can now price a training run on the back of an envelope

What you can price



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- In the exercise you build the envelope